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Optimal capital taxation revisited[☆]

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ABSTRACT

We revisit the question of how capital should be taxed. We allow for a rich set of tax instruments that consists of taxes widely used in practice, including consumption, dividend, capital, and labor income taxes. We restrict policies to those that respect pre-existing promises regarding the current value of wealth. We show that capital should not be taxed (i.e. there should be no intertemporal distortions), if households have preferences that are standard in the macroeconomics literature. We show that Ramsey outcomes that must respect such promises are time consistent. We show that the presumption in the literature that capital should be taxed for some length of time arises because the tax system is restricted.

1. Introduction

How should capital be taxed? An influential literature uses the Ramsey approach in the neoclassical growth model to answer this question.[1] In the Ramsey approach (Ramsey, 1927), the set of tax instruments is exogenously given. Some of this literature (see, e.g., Chamley, 1986; Judd, 1985) considers tax systems in which only labor and capital income can be taxed and the tax rate on capital income is restricted to not exceed an upper bound. This literature finds that capital income should be taxed at its upper bound for some length of time but should not be taxed in the steady state. More recently, Straub and Werning (2015) show that taxing capital income at its upper bound forever may be optimal. This literature leads to the presumption that capital income taxes should be high for some length of time.

In this paper, we take the view that the exogenously given set of tax instruments in the Ramsey approach should include taxes widely used in practice in most developed economies. In addition to taxes on capital and labor income, most economies tax dividends, consumption, and wealth. We refer to a tax system that potentially includes all of these taxes as

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[1] See Judd (1985), Chamley (1986), Zhu (1992), Chari et al. (1994, 1999), Atkeson et al. (1999), Coleman and I. (2000), Bassetto et al. (2006), Werning (2007), and Straub and Werning (2015).

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a *rich tax system*. We also assume that the Ramsey planner cannot reduce the value of initial wealth in utility terms below an exogenously specified level and refer to this constraint on the planner as the *wealth constraint*.[2] The spirit of this wealth constraint is as follows. Before period zero, agents made decisions in expectation of the value of their wealth in period zero, and policies chosen in period zero should not violate those expectations.

As is well known, with a rich tax system, many tax policies can support the same allocations. This multiplicity has led the public finance literature to focus on wedges that the tax system induces in marginal conditions, rather than focusing on the taxes themselves. This multiplicity leads us to focus on whether the Ramsey policy yields intertemporal wedges, rather than focusing on the level of, say, the capital income tax. We say that capital is not taxed if the Ramsey outcome has no intertemporal wedges and that capital is taxed or subsidized depending on the sign of the intertemporal wedge.[3]

We show that with a rich tax system, capital should not be taxed in the steady state of the neoclassical growth model. For general preferences, we show that, along the transition, capital may be taxed or subsidized. We focus attention on a class of preferences that are standard in the macroeconomics literature. These preferences have constant intertemporal elasticity of substitution in consumption and a constant Frisch elasticity in labor. We show that with these preferences, capital should never be taxed. These results hold for any value of the initial wealth that the government is exogenously required to deliver. We also consider environments with uncertainty and show that, with standard macroeconomic preferences, capital should not be taxed.

We show that the presumption in the literature that capital income taxes should be high for some, possibly infinite, length of time arises from restrictions imposed on the tax system and that once we allow for a rich tax system, this presumption disappears. Given that our notion of a rich tax system contains taxes used in most countries, and given that macroeconomic models typically use the preferences we study, our analysis implies that conventional macroeconomic theory strongly suggests that tax systems that distort capital accumulation are inefficient.

One way of thinking about our wealth restriction is that the Ramsey planner is obliged to respect pre-existing promises on the value of initial wealth. Other than this restriction, the planner is free to choose current and future policies. Consider now an alternative environment in which policies are chosen sequentially, with the additional restriction that the planner in each period must respect pre-existing promises regarding the value of inherited wealth. Specifically, suppose that the government in each period must respect promises about the value of wealth made by the government in the previous period and can, in turn, make promises about the value of wealth in the next period in addition to choosing current policies. Other than this promise, the government in the current period has no ability to choose future policies. With this form of partial commitment, we then ask a natural question: What is the equilibrium outcome in an environment in which the government in any period can choose current policies as well as the value of wealth that the government in the following period must respect but has no other form of commitment? We show that the Ramsey outcomes, which have commitment, are Markov equilibrium outcomes in the environment with partial commitment. In this sense, the Ramsey equilibrium with wealth constraints is time consistent. We view this time consistency as a justification for adding wealth constraints to Ramsey problems.

We briefly analyze an economy in which agents are heterogeneous with respect to initial wealth and possibly preferences. This formulation allows for redistributive motives for taxation in addition to the need to raise taxes to finance government spending. With type-specific tax rates, we show that even if preferences are different across agents, it is optimal to not distort capital accumulation. If taxes cannot be type-specific, and if both types of agents have identical standard macro preferences, it is still optimal to never tax capital; that is, there should be no intertemporal distortions (see Werning, 2007, for a similar result without wealth restrictions).

We have argued that neither the need to raise revenues through distorting taxes to finance public goods nor the need to redistribute resources across agents requires intertemporal distortions. If there are other externalities, corrective Pigouvian taxation may be needed, and zero capital taxation may not be optimal. One example of such externalities consists of economies with incomplete markets, and we briefly describe how our methods can be extended to these economies. As we argue, although distorting intertemporal decisions will likely be optimal in a rich tax system, how quantitatively large these distortions should be is still an open question. We develop a simple recursive formulation related to that in Bhandari et al. (2018) which can be used to address these questions. This recursive formulation seems especially appropriate for answering these questions quantitatively.

Our notion of zero taxation of capital is that there are no intertemporal distortions. In general, this notion is not equivalent to production efficiency. We show this lack of equivalence by demonstrating that with general preferences and a rich tax system, the Ramsey allocations are production efficient but do not require zero taxation of capital in the sense of no intertemporal distortions. The Ramsey allocation is production efficient because a rich tax system allows for taxes on all final consumption goods and all types of labor, and allows pure rents to be taxed to meet the wealth constraint. The Ramsey allocation with general preferences typically does not have uniform taxation of consumption goods or labor types. Such uniform taxation is needed to achieve zero taxation of capital in the sense that there are no intertemporal distortions.

With standard macro preferences, however, the optimality of uniform taxation, and therefore of zero taxation of capital, follows from production efficiency. To obtain this result, we recast the growth model as a static model in which the house-

[2] See Armenter (2008) for a similar formulation.
[3] Under this definition, it is possible that both the initial capital in period zero is taxed and dividends are taxed.

hold consumes one final composite consumption good and supplies one composite labor input. Constant returns to scale technologies are used by competitive firms to turn the composite labor input into intermediate goods that consist of labor, consumption, and capital at each date in the original economy. The consumption goods at each date in the original economy are transformed into the composite consumption good by a constant returns to scale technology. The Ramsey planner in the recast economy faces a wealth constraint. We show that not distorting the use of intermediate goods is optimal. This result implies that in the original economy, capital should never be taxed. This result is related to but different from the production efficiency result in Diamond and Mirrlees (1971) in that they require full taxation of pure rents to obtain production efficiency, whereas with our wealth constraint, we do not require such full taxation.

Our result that taxing or subsidizing capital is not optimal clearly conflicts with the general presumption in the literature that taxing capital for some length of time is optimal. The difference between these results arises for two reasons. First, the literature restricts initial policies rather than the value of initial wealth. Second, the literature allows for restricted tax systems that tax only capital and labor income with a cap on capital tax rates, whereas we consider rich tax systems. It turns out, however, that the restriction on initial policies rather than the value of initial wealth plays a relatively minor role in the difference between results. We illustrate this minor role by considering an optimal taxation problem with a rich tax system and restrictions on initial policies. We show that, with standard preferences and a rich tax system, capital is taxed for one period at most and is never taxed after the first period. This result is in stark contrast to the presumption in the literature.

One way of gaining intuition for the presumption in the literature is to begin by noting the well-known result that the Ramsey planner without wealth constraints seeks to completely tax away pure rents. In the growth model, these pure rents consist of the value of pre-existing wealth in utility terms. This value of wealth is the product of the initial marginal utility of consumption and wealth in units of initial consumption goods. A Ramsey planner who cannot directly confiscate wealth in goods terms has a strong incentive to reduce the initial marginal utility of consumption so as to indirectly confiscate the value of wealth. With a restricted tax system that allows taxation of only labor and capital, and with a bound on the capital tax rate, the longer the capital income tax rate is set at its upper bound, the greater the reduction in the initial marginal utility of consumption. The planner trades off the gain from this indirect confiscation with the losses from the induced intertemporal distortions. This trade-off determines the length of time that capital income taxes are set to the upper bound.

The central lesson of the public finance literature stemming from Ramsey (1927) and Diamond and Mirrlees (1971) is that tax systems that include taxes on all final consumption goods and taxes on all primary inputs (such as labor) and tax away all pure rents yield production efficiency. We have extended this result to environments in which the Ramsey planner faces a wealth constraint that limits the ability to fully tax away pure rents. Our notion of zero taxation of capital is stronger than production efficiency but follows from it for the kinds of preferences that are standard in the macroeconomics literature. These observations lead us to conclude that systems that do not allow for a rich tax system may find capital taxation optimal but are of limited interest from an applied perspective, given that most countries already use taxes that constitute a rich tax system.

In this paper we prove some new results, bring together well-known results in a unified framework, clarify the relation between these results, and relate the results on dynamic taxation to those on uniform commodity taxation. The new results in this paper focus on the equivalence between Markov equilibria and commitment equilibria, the insights into the desire to tax capital heavily in restricted tax systems, and the exact relation to the production efficiency literature. For the result that with a rich tax system and standard macro preferences, it is optimal not to tax future capital after a one-period transition, see Chari et al. (1999), Zhu (1992), and Werning (2007). For the result that optimal capital income taxes with a wealth restriction ought to be zero in a steady state, see Armenter (2008). For the heterogeneous agent results with complete markets, see Werning (2007).

2. A representative agent economy

Our benchmark framework is the deterministic neoclassical growth model with taxes. The representative household's preferences are defined over consumption c_t and labor n_t,

$$U = \sum_{t=0}^{\infty} \beta^t u(c_t, n_t), \tag{1}$$

satisfying the usual properties. The production technology is described by

$$c_t + g_t + k_{t+1} - (1-\delta)k_t \le F(n_t, k_t), \tag{2}$$

where k_t is capital, g_t is exogenous government consumption, δ is the depreciation rate, and the production function F is constant returns to scale.

We now describe a competitive equilibrium with taxes. The government finances public consumption and initial debt, b_0, with time-varying proportional taxes. We allow for a rich tax system that includes taxes on consumption τ_t^c, labor income τ_t^n, capital income τ_t^k, dividends τ_t^d, and a tax on initial wealth, l_0.[4]

[4] Note that we allow only for a tax on wealth in period zero. It turns out that allowing for taxes on wealth in future periods is equivalent to a time-varying consumption tax. Since we allow for consumption taxes, taxes on future wealth are redundant.

Capital accumulation is conducted by firms. Given that the technology is constant returns to scale, we assume without loss of generality that the economy has a representative firm. The household owns the firm and receives dividends. We choose this decentralization because it is straightforward to relate the taxes in the model to the ones in existing tax systems.[5] We now describe the household's and firm's problems and define a competitive equilibrium.

Households

The representative household maximizes utility (1), subject to the present-value budget constraint

$$\sum_{t=0}^{\infty} q_t[(1+\tau_t^c)c_t - (1-\tau_t^n)w_t n_t] \le (1-l_0)\left[b_0 + \sum_{t=0}^{\infty} q_t\left(1-\tau_t^d\right)d_t\right], \tag{3}$$

where q_t is the price of one unit of the good produced in period t in units of the good in period zero, so that $q_0 = 1$; w_t is the pretax wage rate; b_0 is the initial holdings of government debt; and d_t are the dividends paid by the firm.

Firms

The representative firm maximizes the after-tax present value of dividends

$$\sum_{t=0}^{\infty} q_t\left(1-\tau_t^d\right)d_t, \tag{4}$$

where dividends, d_t, are given by

$$d_t = F(k_t, n_t) - w_t n_t - \tau_t^k[F(k_t, n_t) - w_t n_t - \delta k_t] - [k_{t+1} - (1-\delta)k_t]. \tag{5}$$

Note that the taxes on capital income, τ_t^k, are levied on income net of depreciation. Note also that the tax on dividends, τ_t^d, effectively allows firms to expense gross investment. This expensing later implies that, as we show below, dividend taxes are similar to consumption taxes in their effect on intertemporal wedges.

In this way of setting up the competitive equilibrium, dividends are net payments to claimants of the firm. These payments could be interpreted either as payments on debt or as payments to equity holders. To clarify this interpretation, consider an all-equity firm. In this case, our notion of dividends consists of cash dividends plus stock buybacks less issues of new equity. In particular, under this interpretation, taxes on capital gains associated with stock buybacks are assumed to be levied on accrual and at the same rate as cash dividends. Note also that dividends could be negative if returns to capital are smaller than investment. In this case, a positive tax on dividends would represent a subsidy to the firm.[6]

Remark: Note that the taxes paid by firms are on accrued profits rather than on imputed profits. With taxation on accrued profits, if, in some period t, $\tau_t^k > 1$, the solution to the firm's problem would not be interior. This observation suggests that it may be reasonable to impose a restriction that $\tau_t^k \le 1$. Similarly, it may be reasonable to impose restrictions on dividend taxes so that the present value of after-tax dividends must be nonnegative. If the taxes are on imputed profits, then it may be possible to have an interior solution without these restrictions. In what follows, we analyze equilibria with and without such restrictions.

Government

The government budget constraint is

$$\sum_{t=0}^{\infty} q_t\left(\tau_t^c c_t + \tau_t^n w_t n_t + \tau_t^d d_t + \tau_t^k[F(k_t, n_t) - w_t n_t - \delta k_t] - g_t\right) + l_0\left[b_0 + \sum_{t=0}^{\infty} q_t\left(1-\tau_t^d\right)d_t\right] = 0.$$

Competitive and Ramsey equilibrium

A *competitive equilibrium* is a set of allocations $\{c_t, n_t, k_{t+1}, d_t\}$, prices $\{q_t, w_t\}$, and policies $\left\{\tau_t^c, \tau_t^n, \tau_t^d, \tau_t^k, l_0\right\}$, given $\{k_0, b_0\}$ such that households maximize utility subject to their constraints, firms maximize the present value of dividends, the government budget constraint is satisfied, and markets clear in that resource constraints (2) are satisfied. We refer to a subset of the allocations $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ as *implementable allocations* if they are part of a competitive equilibrium.

A *Ramsey equilibrium* is a competitive equilibrium that yields the highest utility for the representative household. The *Ramsey allocation* is the associated implementable allocation. To characterize the Ramsey equilibrium, we begin by deriving the conditions that any competitive equilibrium must satisfy. The first-order conditions of the households' problem include

$$-\frac{u_{c,t}}{u_{n,t}} = \frac{(1+\tau_t^c)}{\left(1-\tau_t^n\right)w_t}, \; t \ge 0, \tag{6}$$

$$\frac{u_{c,t}}{1+\tau_t^c} = \frac{q_t}{q_{t+1}}\frac{\beta u_{c,t+1}}{\left(1+\tau_{t+1}^c\right)}, \; t \ge 0, \tag{7}$$

[5] The literature often uses an alternative decentralization in which the households own the capital stock and firms rent capital from the households. In Online Appendix A, we show that the two decentralizations are equivalent.

[6] For an early treatment of dividend taxes see Auerbach (1979). For more recent treatments, see McGrattan and Prescott (2005), Gourio and Miao (2010), and Conesa and Domínguez (2013, 2018, 2019).

where $u_{c,t}$ and $u_{n,t}$ denote the marginal utilities of consumption and labor in period t.

The first-order conditions of the firm's problem include

$$w_t = F_{n,t} \quad \text{and} \tag{8}$$

$$\frac{q_t}{q_{t+1}} = \frac{(1-\tau_{t+1}^d)\left[1+\left(1-\tau_{t+1}^k\right)(F_{k,t+1}-\delta)\right]}{1-\tau_t^d}, \tag{9}$$

where $F_{n,t}$ and $F_{k,t}$ denote the marginal products of capital and labor in period t.

Substituting d_t with (5) and using (8) and (9), it is possible to show that the present discounted value of dividends is given by

$$\sum_{t=0}^{\infty} q_t\left(1-\tau_t^d\right)d_t = \left(1-\tau_0^d\right)\left[1+\left(1-\tau_0^k\right)\left(F_{k,0}-\delta\right)\right]k_0. \tag{10}$$

The budget constraint (3) can then be written as

$$\sum_{t=0}^{\infty} q_t[(1+\tau_t^c)c_t - (1-\tau_t^n)w_t n_t] = W_0, \tag{11}$$

where the initial wealth of households is given by

$$W_0 \equiv (1-l_0)\left[b_0 + \left(1-\tau_0^d\right)\left[1+\left(1-\tau_0^k\right)\left(F_{k,0}-\delta\right)\right]k_0\right]. \tag{12}$$

The full set of equilibrium conditions can then be summarized by the household's first-order marginal conditions (6) and (7), the firm's first-order conditions (8) and (9), the budget constraint (11) with (12), together with the expression for dividends (5) and the market clearing conditions (2). The government's budget constraint is implied by the household budget constraint and market clearing.

Implementability

These equilibrium conditions can be used to provide a compact characterization of the set of implementable allocations. Substituting prices and taxes from the first-order conditions for the households with the households' budget constraint (11), we obtain that any competitive equilibrium must satisfy the following *implementability constraint:*

$$\sum_{t=0}^{\infty} \beta^t (u_{c,t}c_t + u_{n,t}n_t) = \mathcal{W}_0, \tag{13}$$

where

$$\mathcal{W}_0 = \frac{u_{c,0}}{1+\tau_0^c} W_0. \tag{14}$$

Clearly, any competitive equilibrium must also satisfy the resource constraints (2). Thus, we have shown that any competitive equilibrium must satisfy the resource constraints (2) and the implementability constraint (13).

Next we show that given any arbitrary allocation and period zero policies that satisfy (2) and (13), it is possible to construct prices and policies so that these outcomes constitute a competitive equilibrium. Consider one such implementation. Pin down the wage rates w_t from (8). Set the consumption tax rate to zero, $\tau_t^c = 0$ for all $t \geq 1$. Pin down the tax rate on labor τ_t^n from (6). Set the intertemporal prices q_t for $t \geq 1$ from (7). Set $\tau_t^k = 0$ for $t \geq 1$. Given τ_0^d, pin down the time path of dividend taxes, τ_t^d, $t \geq 1$, from (9). Obtain dividends d_t from (5). It is immediate that these allocations satisfy all the equilibrium conditions for households and firms. Thus, the so-constructed allocation, prices, and policies are a competitive equilibrium. We then have the following proposition.

Proposition 1. *(Characterization of the implementable allocations) Any implementable allocation satisfies the resource constraints (2) and the implementability constraint (13). Furthermore, if a sequence* $\{c_t, n_t, k_{t+1}\}$*, initial conditions* k_0, b_0*, and period zero policies* $(\tau_0^c, \tau_0^d, \tau_0^k, l_0)$*, satisfy (2) and (13), it is implementable.*

Wedges and multiple implementations

In proving this proposition, we used one particular implementation of policies. Next we consider other implementations. To see how taxes introduce wedges in marginal conditions, consider the following marginal conditions. These conditions implicitly give an *intratemporal wedge*,

$$-\frac{u_{c,t}}{u_{n,t}} = \frac{(1+\tau_t^c)}{\left(1-\tau_t^n\right)F_{n,t}}, \tag{15}$$

a *consumption intertemporal wedge*,

$$\frac{u_{c,t}}{\beta u_{c,t+1}} = \frac{(1-\tau_{t+1}^d)(1+\tau_t^c)}{\left(1-\tau_t^d\right)\left(1+\tau_{t+1}^c\right)}\left[1+\left(1-\tau_{t+1}^k\right)\left(F_{k,t+1}-\delta\right)\right], \tag{16}$$

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and a *labor intertemporal wedge*,

$$\frac{u_{n,t}}{\beta u_{n,t+1}} = \frac{(1-\tau_{t+1}^d)(1-\tau_t^n)}{\left(1-\tau_t^d\right)\left(1-\tau_{t+1}^n\right)}\frac{F_{n,t}}{F_{n,t+1}}\left[1+\left(1-\tau_{t+1}^k\right)\left(F_{k,t+1}-\delta\right)\right]. \tag{17}$$

Note that the labor intertemporal wedge condition, (17), is implied by (15) and (16). We include it here to analyze when not distorting the labor intertemporal margin is optimal.

Notice that a constant dividend tax does not distort any of the marginal conditions. Such a tax, of course, raises revenues by reducing the value of the firm at the beginning of period zero, which in turn reduces the household's wealth, as can be seen from (12). In this sense, a constant dividend tax is equivalent to a levy on the initial capital stock. This dividend tax resembles the tax proposed by Abel (2007) as a way to collect lump-sum revenue from the taxation of the initial capital stock. Note also that a constant consumption tax does not distort intertemporal conditions but does reduce the value of initial wealth, as can be seen from (14). Notice also that a tax on capital income distorts intertemporal decisions in the same way as do time-varying taxes on consumption, dividends, and labor income. We will use these properties in implementing the Ramsey equilibrium.

To see how a competitive equilibrium can be implemented in multiple ways, consider alternative implementations of some arbitrary competitive equilibrium.

Consider first an alternative implementation that uses a system that levies taxes only on consumption, labor, and initial wealth. We refer to such a system as the *Diamond–Mirrlees system* because it is in the spirit of their tax system that allows taxes only on final goods, primary inputs, and pure rents. Clearly, τ_t^c and τ_t^n can be chosen to satisfy (15)–(17), and l_0 can be chosen to yield the same initial wealth as in the arbitrary equilibrium.

Consider next a version of the implementation used in the proof of Proposition 1 that levies taxes only on labor income and dividends. We refer to this system as the *Abel system* because it resembles the proposal in Abel (2007). Here, τ_t^d and τ_t^n can be chosen to satisfy (15)–(17), and τ_0^d can be chosen to yield the same initial wealth as in the arbitrary equilibrium. This implementation may require τ_0^d to be greater than 100%. So if we impose the restriction that $\tau_t^d \leq 1$, implementing some competitive equilibria may not be possible.

Finally, consider an alternative implementation that uses taxes only on labor and capital income, referred to as the *Chamley–Judd system*. Again, τ_t^n and τ_{t+1}^k can clearly be chosen to satisfy (15)–(17), and τ_0^k in the alternative implementation can be chosen to yield the same initial wealth as in the arbitrary equilibrium. Analogously to the Abel implementation, note that in this implementation, the tax rates on capital income may need to be greater than one. So if we impose the restriction that $\tau_t^k \leq 1$, implementing some competitive equilibria may not be possible.

Intertemporal distortions

We now turn to the question of whether capital should be taxed. Given our results on multiple implementations, it is clear that setting capital income taxes to zero does not mean that the economy has no intertemporal wedges. We will say that capital is not taxed if the Ramsey allocation has no intertemporal wedges. Formally, a competitive equilibrium has *no intertemporal distortions in consumption* from period s onward if there is no wedge in (16) in that

$$\frac{u_{c,t}}{\beta u_{c,t+1}} = 1 + F_{k,t+1} - \delta, \text{ for all } t \geq s. \tag{18}$$

Similarly, a competitive equilibrium has *no intertemporal distortions in labor* from period s onward if there is no wedge in (17) if

$$\frac{u_{n,t}}{\beta u_{n,t+1}} = \frac{F_{n,t}}{F_{n,t+1}}\left(1 + F_{k,t+1} - \delta\right) \text{ for all } t \geq s. \tag{19}$$

Finally, a competitive equilibrium has *no taxation of capital* from period s onward if (18) and (19) hold.

Note that it follows from (16) and (17) that no taxation of capital implies constant intratemporal distortions in (15).

2.1. Ramsey equilibrium

We assume that the Ramsey planner faces a wealth constraint in the sense that households must be allowed to keep an exogenous value of initial wealth $\bar{\mathcal{W}}$, measured in units of utility. Specifically, we impose the following *wealth restriction in utility terms* on the Ramsey problem:

$$\mathcal{W}_0 \geq \bar{\mathcal{W}}. \tag{20}$$

With this restriction, policies, including initial policies, can be chosen arbitrarily, but households must receive a value of initial wealth in utility terms of $\bar{\mathcal{W}}$ (see Armenter, 2008, for an analysis with such a restriction). The spirit of this restriction is that before period zero, agents made decisions in expectation of the value of their wealth in period zero, and policies chosen in period zero should not violate those expectations. We make this idea precise in the section on partial commitment below.

Given Proposition 1, it follows that the Ramsey allocation, together with period zero policies, maximizes utility subject to (2), (13), and (20). We now characterize the first-order necessary conditions for an interior solution to the Ramsey problem, as follows:

$$-\frac{u_{c,t}}{u_{n,t}} = \frac{1+\varphi\left(1+\sigma_t^{n}-\sigma_t^{nc}\right)}{1+\varphi\left(1-\sigma_t-\sigma_t^{cn}\right)}\frac{1}{F_{nt}},\ t\geq 0, \tag{21}$$

$$\frac{u_{c,t}}{\beta u_{c,t+1}} = \frac{1+\varphi\left(1-\sigma_{t+1}-\sigma_{t+1}^{cn}\right)}{1+\varphi\left(1-\sigma_t-\sigma_t^{cn}\right)}\left(1+F_{k,t+1}-\delta\right),\ t\geq 0, \tag{22}$$

implying

$$\frac{u_{n,t}}{\beta u_{n,t+1}} = \frac{1+\varphi\left(1+\sigma_{t+1}^{n}-\sigma_{t+1}^{nc}\right)}{1+\varphi\left(1+\sigma_t^{n}-\sigma_t^{nc}\right)}\frac{F_{n,t}\left(1+F_{k,t+1}-\delta\right)}{F_{n,t+1}},\ t\geq 0, \tag{23}$$

together with the implementability and resource constraints. Here,

$$\sigma_t = -\frac{u_{cc,t}c_t}{u_{c,t}},\ \sigma_t^{n} = \frac{u_{nn,t}n_t}{u_{n,t}},\ \sigma_t^{nc} = -\frac{u_{nc,t}c_t}{u_{n,t}},\ \sigma_t^{cn} = -\frac{u_{cn,t}n_t}{u_{c,t}},$$

and φ is the multiplier of the implementability condition.

These conditions make it clear that the optimal wedges depend on their own and cross elasticities of consumption and labor. If those elasticities are constant, not having intertemporal distortions is optimal. In this case, intratemporal wedges are constant and generally positive. If the elasticities are not constant over time, having intertemporal distortions is optimal, but whether effectively taxing or subsidizing capital accumulation is optimal depends on whether elasticities are increasing or decreasing over time.

Note that if consumption and labor are constant over time, then the relevant elasticities are also constant, so that it is optimal to have no intertemporal distortions. This observation leads to the following well-known proposition.

Proposition 2. *(No intertemporal distortions in the steady state) If the Ramsey equilibrium converges to a steady state, not having intertemporal distortions is optimal asymptotically.*

Now consider preferences that are standard in the macroeconomics literature. These preferences take the form

$$U = \sum_{t=0}^{\infty}\beta^t\left(\frac{c_t^{1-\sigma}-1}{1-\sigma}-\eta n_t^{1+\psi}\right). \tag{24}$$

In this case, the elasticities are constant, so that we have the following proposition.

Proposition 3. *(No intertemporal distortions ever) Suppose that preferences are given by (24) and that the wealth restriction (20) must be satisfied. The Ramsey solution then has no intertemporal distortions for all $t\geq 0$.*

Proposition 3 extends straightforwardly to environments with uncertainty. To show this result, in Online Appendix B we prove an analog of Proposition 3 in a stochastic model with fluctuations in government spending and technology.

Remark. For general preferences, proving that the economy converges to a steady state is difficult. For standard preferences, proving that it does so is straightforward.

Note that the preferences above are separable and homothetic in both consumption and labor. (In Appendix C, we show that they are the only time-separable preferences with those properties.) We use these properties in Section 4 to relate our results to those on uniform commodity taxation and production efficiency.

The Ramsey outcomes characterized in Proposition 3 can be implemented with a variety of systems. Each of these systems is a restricted version of our rich tax system. Some of these restricted tax systems allow for implementation of our Ramsey equilibrium for any initial conditions, whereas others allow for implementation for only a particular set of initial conditions. The Diamond-Mirrlees system, which allows for taxes on consumption, labor, and initial wealth, can implement the Ramsey equilibrium for any initial conditions. For example, one implementation has constant tax rates on consumption, sets the initial wealth tax to satisfy the wealth constraint, and sets the labor tax to zero, whereas another implementation has constant tax rates on labor, sets the initial wealth tax appropriately, and sets consumption taxes to zero.

Next we consider systems that allow for implementation for only a subset of initial conditions. Consider first a Diamond-Mirrlees system with a zero wealth tax. If W_0 and $\bar{\mathcal{W}}$ are of the same sign, then a consumption tax acts exactly like a wealth tax, so that such a system can implement the Ramsey outcome. For example, a constant consumption tax could be set so as to have the same effect as the wealth tax, and the labor tax could be set appropriately to satisfy the intratemporal condition. If W_0 and $\bar{\mathcal{W}}$ are of different signs, then a wealth tax greater than one can implement the Ramsey equilibrium, whereas the consumption tax cannot. Thus, this system implements the Ramsey outcome only for a subset of initial conditions.

Consider next an Abel system. Again, if W_0 and $\bar{\mathcal{W}}$ are of the same sign, an Abel system with an unrestricted dividend tax can implement the Ramsey outcome because the dividend tax is similar to the wealth tax. As with the Diamond and Mirrlees implementation, a constant dividend tax could be set so as to have the same effect as the wealth tax, and the labor tax could be set to satisfy the intratemporal condition. If dividend taxes are restricted to be less than 100%, then the Abel system may not be able to implement the Ramsey allocation. To see this, let W_0 and τ_0^c denote the initial wealth in units of goods and the initial tax rate on consumption in the Diamond-Mirrlees implementation without a wealth tax. The Abel system then implements the Ramsey outcome if and only if the following condition is met:

$$\frac{W_0}{1+\tau_0^c} \geq b_0. \tag{25}$$

Note that $W_0 > b_0$, so that the Abel system implements the Ramsey outcome if the effective tax on initial wealth arising from the tax on consumption τ_0^c is not too large.

As with the Abel system, the Chamley–Judd system can implement the Ramsey outcome if the capital income tax is unrestricted. Here, the labor tax implements the intratemporal wedge, and if $F_{k,0} - \delta > 0$, the initial capital income tax can be used to satisfy the initial wealth constraint. If capital income taxes are restricted to be less than 100%, then this system can implement the Ramsey outcome if and only if the following condition is met:

$$\frac{W_0}{1+\tau_0^c} \geq b_0 + k_0. \tag{26}$$

Notice that condition (25) is stronger than condition (26). The reason for this difference is that the dividend taxes are levied on both the period zero net capital income and the value of the capital, whereas the capital income tax is levied only on the period zero net capital income.

Debt of multiple maturities

Next we analyze Ramsey equilibria when the inherited debt has longer maturities than the single maturity debt that we have assumed so far. Let b_t^0 be the amount of debt inherited in period 0 that matures in period t. Initial wealth is then given by

$$W_0 = (1-l_0)\left[\sum_{t=0}^{\infty} q_t b_t^0 + \left(1-\tau_0^d\right)\left[k_0 + \left(1-\tau_0^k\right)\left(F_{k,0}-\delta\right)k_0\right]\right].$$

The implementability condition is the same as (13), with

$$\mathcal{W}_0 = (1-l_0)\left[\sum_{t=0}^{\infty} \frac{\beta^t u_{c,t}}{\left(1+\tau_t^c\right)} b_t^0 + \frac{u_{c,0}\left(1-\tau_0^d\right)}{\left(1+\tau_0^c\right)}\left[1+\left(1-\tau_0^k\right)\left(F_{k,0}-\delta\right)\right]k_0\right].$$

Clearly, the solution to the Ramsey problem is the same as in the model with one-period debt. Note that the same reasoning applies if the government is committed to lump-sum transfers in future periods.

2.2. Ramsey equilibria with restrictions on taxes

Here we relate our results to an extensive and influential literature. This literature differs from our analysis in two ways. First, the literature typically imposes restrictions on initial policies, as opposed to our wealth restriction. Second, it considers tax systems that are more restricted than our rich tax system. For example, Chamley (1986), Judd (1985), and Straub and Werning (2015) consider systems in which the only taxes allowed are taxes on capital and labor income and in which the tax rate on capital is restricted to be below an upper bound, typically 100% in both the initial period and subsequent periods. This literature finds that the optimal tax rate on capital income is at its upper bound for some length of time. Straub and Werning (2015) show that the tax rate on capital can be at its upper bound forever.[7] Both types of restrictions play a role in the results in the literature, but it turns out that restricted tax systems play a much more important role than restrictions on initial policies.

Consider first a rich tax system with restrictions on initial policies. Specifically, we assume that l_0, τ_0^d, τ_0^k, and τ_0^c are exogenously given.[8] The implementability constraint becomes

$$\sum_{t=0}^{\infty} \beta^t (u_{c,t}c_t + u_{n,t}n_t) = \frac{u_{c,0}(1-l_0)}{1+\tau_0^c}\left[b_0 + \left(1-\tau_0^d\right)\left[1+\left(1-\tau_0^k\right)\left(F_{k,0}-\delta\right)\right]k_0\right]. \tag{27}$$

The Ramsey problem is to maximize utility subject to (2) and (27). The first-order conditions of the Ramsey problem are the same as before in (21), (22), and (23), for all $t \geq 1$. The other first-order conditions for period zero are different from those in our benchmark problem. The intertemporal condition for consumption between periods zero and one is now

$$\frac{u_{c,0}}{\beta u_{c,1}} = \frac{1+\varphi\left(1-\sigma_1+\sigma_1^{cn}\right)}{1+\varphi\left(1-\sigma_0+\sigma_0^{cn}+\frac{\sigma_0 \mathcal{V}}{c_0}\right)}\left(1+F_{k,1}-\delta\right), \tag{28}$$

for $\mathcal{V} = \left((1-l_0)/\left(1+\tau_0^c\right)\right)\left[b_0 + \left(1-\tau_0^d\right)\left[1+\left(1-\tau_0^k\right)\left(F_{k,0}-\delta\right)\right]k_0\right]$. We omit the intratemporal condition for period zero since it is not used in deriving our main results.

[7] Bassetto et al. (2006) obtain a similar result in a political economy model.

[8] Alternatively, we could have assumed that each of these tax rates has an upper bound, in which case the solution would be to trivially set them equal to their upper bounds.

With standard macro preferences, since elasticities are constant over time, not having intertemporal distortions from period one onward is optimal. Now consider intertemporal distortions in period zero. With standard macro preferences, $\sigma_1 = \sigma_0$ and cross elasticities are zero, so that if $\mathcal{V} > 0$, (28) implies that

$$\frac{u_{c,0}}{\beta u_{c,1}} < 1 + F_{k,1} - \delta. \tag{29}$$

Comparing (16) with (29), we see that the effective implied tax rate on capital income in period one is strictly positive. One intuition for this result is as follows. The Ramsey planner finds it optimal to reduce the right side of the implementability constraint (27) or, equivalently, the value of the household wealth in utility terms. This value can be reduced by decreasing the marginal utility of period zero consumption. We summarize this discussion in the following proposition.

Proposition 4. *(No intertemporal distortions after one period) Suppose preferences satisfy (24) and initial policies are exogenously specified, with no wealth restriction. The Ramsey solution then has no intertemporal distortions for all $t \geq 1$. If $\mathcal{V} > 0$, effectively taxing capital accumulation from period zero to period one is optimal.*

As usual, the Ramsey outcome can be implemented in a variety of ways. One implementation uses dividend and labor income taxes alone, together with the exogenously given initial policies. In this implementation, labor income taxes are set to satisfy the intratemporal wedge condition (15). Note that, for $t \geq 1$, the labor income tax rate is constant. Dividend taxes are set to satisfy the intertemporal wedge condition (16). From this condition, it is clear that the dividend tax is below one in period one and zero thereafter.

An alternative implementation uses consumption and labor income taxes alone. Consumption taxes are set to satisfy the intertemporal wedge condition (16), and they are constant starting in period one. Given these consumption taxes, labor income taxes are set to satisfy (15). Inspecting (16) and (29), we clearly see that the consumption tax rate in period one, τ_1^c, must be greater than the tax rate in period 0, τ_0^c. Indeed, τ_1^c could be so large that the associated labor income tax rate needed to satisfy the intratemporal condition (15) might be negative. This possibility may be a disadvantage for a consumption tax implementation. Note that the dividend tax implementation does not have this disadvantage.

The third implementation uses labor and capital income taxes alone, together with the exogenously given initial policies. This implementation is the one widely used in the literature. As in the other implementations, the labor tax is used to satisfy (15). The capital income tax rate is used to satisfy (16). Note that, for $t \geq 2$, the capital income tax rate is zero. Inspecting (16) and (29), we see that the capital income tax rate in period one, τ_1^k, could be greater than 100%. The dividend tax is bounded below one with the dividend tax implementation, but the capital income tax may be greater than one in this implementation because the dividend tax is a tax on the gross return on capital, whereas the capital income tax is a tax on the net return.

These three implementations show that, with a rich tax system, intertemporal decisions are distorted for one period at most. This finding implies that the results in the literature arise not just from restrictions on initial policies but also from departures from a rich tax system.

To understand the role of restrictions on the tax system, consider a tax system that is restricted in that only capital and labor income can be taxed and the tax rate on capital income must be below an exogenously specified level $\bar{\tau} \leq 1$. Rearranging (16), it is immediate that this restriction imposes additional constraints on the Ramsey problem given by

$$\frac{u_{c,t}/\beta u_{c,t+1} - 1}{F_{k,t+1} - \delta} \geq 1 - \bar{\tau}, \text{ for all } t. \tag{30}$$

In this case, the Ramsey problem is to choose allocations $\{c_t, n_t, k_{t+1}\}$ and τ_0^k to maximize utility subject to the resource constraints (2), the implementability constraint (27), and (30). Since only capital income and labor can be taxed, in (27) we set $l_0 = \tau_0^c = \tau_0^d = 0$.

The constraint (30) may bind for a finite number of periods as in Chamley (1986) or forever as in Straub and Werning (2015). Straub and Werning (2015) set $\bar{\tau}$ to be 100% and show that the optimal solution for particularly high levels of initial debt may be to have the capital income tax set at 100% forever.

To obtain some intuition for these results, notice that the planner has an incentive to make $u_{c,0}$ small so as to reduce the right side of the implementability constraint, (27). Since this right side can also be reduced by confiscating capital, we refer to this incentive as the *confiscation motive*. In determining the optimal tax rates, the planner trades off the gains from the confiscation motive against the losses from intertemporal distortions. To understand this trade-off, consider the intertemporal condition

$$\frac{u_{c,t}}{\beta u_{c,t+1}} = 1 + \left(1 - \tau_{t+1}^k\right)(F_{k,t+1} - \delta).$$

Given $u_{c,1}$, the confiscation motive provides an incentive to make τ_1^k large to reduce $u_{c,0}$. If the confiscation motive is sufficiently strong, the bound on τ_1^k is met. In this case, the planner has an incentive to make $u_{c,1}$ small to further reduce $u_{c,0}$, thereby confiscating initial wealth to a greater extent. Fixing $u_{c,2}$, $u_{c,1}$ in turn can be made small by making τ_2^k large. Again, if the confiscation motive is sufficiently strong, the upper bound will be met. This recursion suggests that the Ramsey solution will have capital taxes be at the upper bound for a length of time. If the initial debt is sufficiently large, the confiscation motive is very strong, and the length of time could be infinite, as pointed out by Straub and Werning (2015).

With, say, dividend taxes, it is possible to reduce $u_{c,0}$ relative to $u_{c,1}$ to an arbitrary extent without distorting intertemporal decisions from period one onward. That is, the after-tax interest rate between period zero and period one can be made negative. With capital income taxes bounded by 100%, $u_{c,t}$ can be reduced relative to $u_{c,t+1}$ only to a limited extent. That is, the after-tax interest rate between any two periods can be reduced to zero at most. The confiscation motive makes it desirable to flatten the entire term structure to zero. If this motive is sufficiently strong, then capital taxes will be 100% forever.

In sum, the results in the literature arise from restrictions on the tax system. These restrictions exclude a multitude of commonly used taxes.

2.3. Partial commitment equilibria

The notion of a Ramsey equilibrium is developed in an environment in which the government commits to an infinite sequence of policies in period zero. Here we consider an alternative institutional framework in which the government has partial commitment. We develop a notion of equilibrium for such an environment, referred to as a *partial commitment equilibrium.* In this environment, in any period, governments lack full commitment in that they cannot specify the entire sequence of policies that will be chosen in the future. They do have the ability to constrain the set of policies in the subsequent period. We first consider constraints on the one-period-ahead value of wealth in utility terms. We then consider constraints on one-period-ahead policies.

Partial commitment to value of wealth

We begin by showing that the Ramsey problem can be written in a recursive form. To do so, we assume throughout the limiting condition $lim_{T\to\infty}\beta^T\mathcal{W}_{T+1}=0$. Under this assumption, the implementability condition can be written equivalently as a sequence of implementability constraints of the form

$$\beta\mathcal{W}_{t+1}+u_{c,t}c_t+u_{n,t}n_t=\mathcal{W}_t. \tag{31}$$

Now the Ramsey problem is to maximize utility (1), subject to the sequence of implementability constraints (31) and the resource constraints. Standard dynamic programming arguments, as in Stokey et al. (1989), imply that this Ramsey problem can be written recursively as

$$V(k,\mathcal{W})=\text{Max } u(c,n)+\beta V(k',\mathcal{W}') \tag{32}$$

subject to

$$c+g+k'-(1-\delta)k\le F(n,k) \text{ and} \tag{33}$$

$$\beta\mathcal{W}'+u_c c+u_n n=\mathcal{W}, \tag{34}$$

where, for simplicity, we have assumed that government consumption is constant, $g_t=g$.

Consider now an environment with partial commitment in that the government in each period chooses current policies and the value of wealth in utility terms for the following period. The government in the current period must respect the value of wealth that the previous government has chosen. We develop a notion of a Markov equilibrium with partial commitment, which we call a *nonconfiscatory equilibrium*. The state of the economy in period t is given by $s=\{k,\mathcal{W}\}$.

Let $\hat{V}(s')$ denote the continuation value induced by the choices of the government in future periods. The government's problem is to solve

$$\hat{V}(k,\mathcal{W})=\text{Max } u(c,n)+\beta\hat{V}(k',\mathcal{W}') \tag{35}$$

subject to (33) and (34). Let $\hat{h}(s)$ denote the solution to (35).

A Markov equilibrium with partial commitment, a nonconfiscatory equilibrium, consists of value functions $\hat{V}(s)$ and policy functions $\hat{h}(s)$, which solve (35) for all s.

Clearly, $\hat{V}(s)=V(s)$, so that we have the following proposition.

Proposition 5. *(Partial commitment is full commitment) The Ramsey equilibrium with wealth restrictions is a Markov equilibrium with partial commitment.*

In our view, an attractive feature of these results is that even if governments in the previous periods have, for whatever reason, not pursued Ramsey policies, current governments will follow Ramsey policies as long as they believe future governments will do so as well.

In Online Appendix F, we consider an alternative environment with partial commitment to returns to establish equivalence between Ramsey and Markov equilibria. This alternative environment builds on Kydland and Prescott (1980) method for computing Ramsey outcomes. They show that a Ramsey equilibrium could be characterized recursively starting in period one, with the addition of a state variable. This state variable represents promised marginal utilities or, alternatively, returns. The government in period zero maximizes discounted utility while being unconstrained by the added state variable. An extensive literature has exploited this recursive formulation to characterize commitment outcomes. We show here that their insight can be used to prove that equilibria in environments in which policymakers have partial commitment to the value of wealth, or equivalently returns, coincide with equilibria with full commitment and initial wealth constraints.

Partial commitment to instruments

Some of the Ramsey literature, starting with Lucas and Stokey (1983) and Chari et al. (1999), restricts period zero policies, as opposed to our restrictions on period zero wealth in utility terms. In environments with partial commitment, one interpretation of this formulation is that governments can commit to policies one period ahead. Partial commitment to policies alone typically cannot implement the Ramsey outcomes. Specifically, consider an alternative form of partial commitment in which the government in period t chooses a subset of policies, $\left\{\tau^c_{t+1}, \tau^k_{t+1}, \tau^d_{t+1}\right\}$, that will be implemented in period $t+1$. The government in any period t is free to choose the labor income tax, τ^n_t. In Chari et al. (2018), we formally define Markov equilibria for this environment. Here we simply note that an extensive literature has shown that Markov and Ramsey equilibria typically do not coincide for environments like the ones considered here.

Chari and Kehoe (1993) characterize Markov equilibria in the Lucas and Stokey (1983) environment and show that Markov and Ramsey equilibria do not coincide. Klein et al. (2008) characterize Markov equilibria in environments similar to ours, with partial commitment to instruments, and show that these equilibria do not coincide with Ramsey equilibria. The results in these papers imply that Markov outcomes are in general different from commitment outcomes. Together with our results on partial commitment on wealth, this result shows that the nature of partial commitment plays a crucial role in determining whether Markov equilibria coincide with commitment equilibria.

3. Heterogeneous agents

Here we briefly discuss extending our results to heterogeneous agents models, as in Judd (1985). The analysis here is closely related to that in Werning (2007). This extension allows us to consider redistributive motives for taxation as well as the need to raise funds to finance public goods. We only consider standard macro preferences and wealth constraints. If the parameters of preferences are the same across agents, our result that it is optimal to not tax capital, in the sense of no intertemporal distortions, continues to hold. If they are not the same and taxes cannot be type-specific, it may be optimal to introduce intertemporal distortions. If taxes can be type-specific, zero capital taxation is optimal even if preferences differ across agents.

We consider an economy with an equal measure of two types of agents, 1 and 2. The social welfare function is $\theta U^1 + (1-\theta)U^2$, with weight $\theta \in [0, 1]$. The individual preferences are assumed to be the standard preferences allowing for possibly different elasticities for the two types of agents,

$$U = \sum_{t=0}^{\infty} \beta^t \left[\frac{\left(c^i_t\right)^{1-\sigma^i} - 1}{1-\sigma^i} - \eta^i \left(n^i_t\right)^{1+\psi^i} \right].$$

The resource constraints are

$$c^1_t + c^2_t + g_t + k_{t+1} - (1-\delta)k_t \le A_t F\left(n^1_t + n^2_t, k_t\right),$$

where $k_t = k^1_t + k^2_t$.

The taxes are the ones in the rich tax system considered in the representative agent economy that includes taxes on consumption τ^c_t, labor income τ^n_t, capital income τ^k_t, dividends τ^d_t, and a tax on initial wealth, l_0. We assume for now that agents cannot be taxed differently based on type.

The implementability conditions can be written as

$$\sum_{t=0}^{\infty} \beta^t \left(u^i_{c,t} c^i_t + u^i_{n,t} n^i_t \right) = \bar{\mathcal{W}}^i, \tag{36}$$

for $i = 1, 2$, where $\bar{\mathcal{W}}^i$ denotes the wealth restriction of agent i.[9]

Since the taxes must be the same for the two agents, an implementable allocation must also satisfy the following marginal conditions:

$$\frac{u^1_{c,t}}{u^2_{c,t}} = \frac{u^1_{n,t}}{u^2_{n,t}} \text{ and } \frac{u^1_{c,t}}{u^2_{c,t}} = \frac{u^1_{c,t+1}}{u^2_{c,t+1}}.$$

These conditions can be written as

$$u^1_{c,t} = \gamma u^2_{c,t} \text{ and } u^1_{n,t} = \gamma u^2_{n,t}, \tag{37}$$

where γ is some endogenous number.[10]

[9] We assume that the feasible set is non-empty. This assumption is satisfied only for some pairs of exogenously specified wealth.

[10] See also Greulichy et al. (2016). Werning (2007) also computes optimal taxes with heterogeneous agents taking advantage of this proportionality of marginal utilities.

Let φ^1 and φ^2 be the multipliers of the two implementability conditions, (36) for $i = 1, 2$, and let λ_t be the multiplier of the resource constraint. The first-order conditions for $t \geq 0$ imply[11]

$$u^2_{c,t} \frac{\gamma\left[\theta + \varphi^1\left(1 - \sigma^1\right)\right]\frac{\sigma^2}{c^2_t} + \left[(1-\theta) + \varphi^2\left(1 - \sigma^2\right)\right]\frac{\sigma^1}{c^1_t}}{\frac{\sigma^2}{c^2_t} + \frac{\sigma^1}{c^1_t}} = \lambda_t \text{ and} \tag{38}$$

$$u^2_{n,t} \frac{\gamma\left[\theta + \varphi^1\left(1 + \psi^1\right)\right]\frac{\psi^2}{n^2_t} + \left[(1-\theta) + \varphi^2\left(1 + \psi^2\right)\right]\frac{\psi^1}{n^1_t}}{\frac{\psi^2}{n^2_t} + \frac{\psi^1}{n^1_t}} = -\lambda_t F_{n,t}, \tag{39}$$

which, together with

$$-\lambda_t + \beta\lambda_{t+1}\left(F_{k,t+1} + 1 - \delta\right) = 0, \tag{40}$$

imply that, if elasticities are equal, $\sigma^1 = \sigma^2 = \sigma$ and $\psi^1 = \psi^2 = \psi$, capital should never be taxed. To see this, notice that, from (37), c^1_t must be proportionate to c^2_t, $c^1_t = \gamma^{-\frac{1}{\sigma}} c^2_t$, and n^1_t must also be proportionate to n^2_t, $n^1_t = (\gamma)^{\frac{1}{\psi}} n^2_t$. It then follows that the terms multiplying the marginal utilities on the left side of (38) and (39) are time invariant.

Note that if the elasticities were to differ across the different agents, the allocations would not be proportionate and the result would not hold. In this case, imposing intertemporal distortions would be optimal. The reason behind this result will become apparent in Section 4 on the relation of our results to the optimality of production efficiency in Diamond and Mirrlees (1971).

We summarize the discussion in the following proposition.

Proposition 6. *(Intertemporal distortions in heterogeneous agent economies) Suppose that preferences for all types of agents are in the class of standard macroeconomic preferences, as in (24). If all agents have the same preferences, then the Ramsey equilibrium has no intertemporal distortions for all $t \geq 0$. If the preferences are different in that $\sigma^1 \neq \sigma^2$ or $\psi^1 \neq \psi^2$, then the Ramsey equilibrium has intertemporal distortions.*

This proposition shows that with standard and identical preferences, allowing for redistributive concerns does not overturn the result that, with a rich tax system, capital should not be taxed. If the preferences of agents are different, then intertemporal distortions are typically optimal. If we allow for type-specific tax rates, then we drop (37) as a constraint on the Ramsey problem and showing that capital should not be taxed is straightforward. In Section 4, we relate all these results to those in the literature on uniform taxation and production efficiency.

3.1. An economy with idiosyncratic shocks

Here, we briefly describe how our methods can be extended to economies with incomplete markets. In such economies, Aiyagari (1995) has argued that taxing capital is optimal (see also Aiyagari and McGrattan, 1998).[12] A recent literature (see, e.g., Conesa et al., 2009; Heathcote et al., 2017; Dyrda and Pedroni, 2018; Açikgöz et al., 2018; and Chien and Wen, 2019) has examined Ramsey problems in incomplete markets with restricted tax systems. The results for the optimal tax systems are quite different across these papers. Although distorting intertemporal decisions will likely be optimal in a rich tax system, how quantitatively large these distortions should be is still an open question. Here, we develop a simple recursive formulation related to that in Bhandari et al. (2018), which can be used to address these questions. We show that market incompleteness introduces additional restrictions on the set of implementable allocations. These additional restrictions imply that our proofs do not apply. Developing quantitative answers to these questions is beyond the scope of this paper, but the recursive formulation seems especially appropriate for these purposes.

We consider an economy with a continuum of ex ante identical agents. Each agent is subject to idiosyncratic shocks to productivity. These shocks have finite support and are i.i.d. across agents The effective units of labor supplied by an agent of type i with shock θ_t in period t is given by $\theta_t n_t$, where n_t are the raw units of labor supplied by the agent. Let θ^t denote the history of shocks, and let $\pi(\theta^t)$ denote the probability of a history of shocks. We assume that agents with the same history of shocks are treated in the same way. An allocation consists of $c_t(\theta^t)$, $n_t(\theta^t)$, k_t for all histories.

The resource constraints in this economy are

$$\sum_{\theta^t} \pi\left(\theta^t\right) c_t\left(\theta^t\right) + g + k_{t+1} - (1-\delta)k_t \leq A_t F\left(\sum_{\theta^t} \pi\left(\theta^t\right)\theta_t n_t\left(\theta^t\right), k_t\right). \tag{41}$$

The individual preferences are

$$U = \sum_{t=0}^{\infty} \sum_{\theta^t} \beta^t \pi\left(\theta^t\right) u\left(c_t\left(\theta^t\right), n_t\left(\theta^t\right)\right). \tag{42}$$

[11] See Online Appendix D for the derivation.

[12] The correct notion of optimality in incomplete markets models is controversial. See Dávila et al. (2012) for a discussion.

We begin by assuming that markets are complete and that agents can be taxed differently based on the realizations of their shocks. We show that the results with complete markets are essentially identical to those in the representative agent model. The complete markets model helps to set the stage for the incomplete markets model, which is the main focus of this section. We consider a rich tax system that includes taxes on consumption $\tau_t^c\left(\theta^t\right)$, labor income $\tau_t^n\left(\theta^t\right)$, capital income τ_t^k, dividends τ_t^d, and a tax on initial wealth, l_0. We show that with standard macro preferences, the Ramsey outcome has wedges that are independent of the realization of the shocks and are constant over time. It follows that, if markets are complete, imposing the restriction that taxes cannot depend on the realization of the shocks does not change the Ramsey outcome.

The budget constraints are[13]

$$\sum_{t=0}^{\infty} q_t\left(\theta^t\right)\left[\left(1+\tau_t^c\left(\theta^t\right)\right)c_t\left(\theta^t\right)-\left(1-\tau_t^n\left(\theta^t\right)\right)\theta_t w_t n_t\left(\theta^t\right)\right]=W_0,$$

where $q_t(\theta^t)$ is the price of one unit of consumption at history θ^t, and where the initial wealth of the households is given by $W_0 \equiv (1-l_0)\left[b_0+\left(1-\tau_0^d\right)\left[1+\left(1-\tau_0^k\right)\left(F_{k,0}-\delta\right)\right]k_0\right]$.

Using the marginal conditions of households, the implementability conditions can be written as

$$\sum_{t=0}^{\infty}\sum_{\theta^t}\beta^t\pi\left(\theta^t\right)\left(u_{c,t}\left(\theta^t\right)c_t\left(\theta^t\right)+u_{n,t}\left(\theta^t\right)n_t\left(\theta^t\right)\right)=\frac{u_{c,0}(\theta)W_0}{1+\tau_0^c(\theta)}=\bar{\mathcal{W}}, \tag{43}$$

where $\bar{\mathcal{W}}$ denotes the wealth restriction.

The Ramsey problem in this economy is to maximize (42), subject to resource constraints, (41), and implementability condition (43). In Online Appendix E, we prove the following proposition.

Proposition 7. *If agents have standard macro preferences, the Ramsey outcome has intratemporal distortions that are independent of the history of shocks and are constant over time, so that there are no intertemporal distortions.*

This proposition implies that, with complete markets, imposing the restriction that tax rates should be the same for all agents in any period does not change the Ramsey outcome.

We now develop a recursive formulation of this problem. This recursive formulation is convenient when we consider the Ramsey problem with incomplete markets below. For simplicity we assume that the shocks are i.i.d. over time, and we let $\pi(\theta)$ denote the probability of shock $\theta_t=\theta$. It is straightforward to show that the recursive formulation can be modified to handle economies in which the shock follows a Markov process. Let $\mathcal{W}_t\left(\theta^{t-1}\right)$ denote the *ex ante promised wealth* of an agent in utility terms in period t before the current shock is realized,

$$\mathcal{W}_t\left(\theta^{t-1}\right)=\sum_{\theta}\pi(\theta)\frac{u_{c,t}\left(\theta^t\right)W_t\left(\theta^t\right)}{1+\tau_t^c(\theta^t)},$$

where $W_t(\theta^t)$ denotes the wealth in (before-tax) consumption units at the beginning of period t. The single implementability constraint (36) can then be written as a sequence of implementability constraints of the form

$$\sum_{\theta}\pi(\theta)\left[\beta\mathcal{W}_{t+1}\left(\theta^t\right)+u_{c,t}\left(\theta^t\right)c_t\left(\theta^t\right)+u_{n,t}\left(\theta^t\right)n_t\left(\theta^t\right)\right]=\mathcal{W}_t\left(\theta^{t-1}\right). \tag{44}$$

We impose the restriction that $\lim_{T\to\infty}\beta^T\mathcal{W}_T\left(\theta^{T-1}\right)=0$, a.s. Under this restriction, (36) and (44) are equivalent.

This formulation of the implementability constraints suggests a recursive formulation in which, rather than recording the history θ^{t-1} for each agent, we simply record the ex ante promised wealth $\mathcal{W}$ for that agent. The aggregate state variables, then, consist of a cumulative distribution H over ex ante promised wealth, $\mathcal{W}$, and the capital stock, k.

The Ramsey problem can be written recursively as choosing $c(\theta,\mathcal{W})$, $n(\theta,\mathcal{W})$, k', the policy function for future promised wealth $\mathcal{W}'=g(\theta,\mathcal{W})$, and the implied distribution over ex ante promised wealth H', to solve

$$V(k,H)=\text{Max}\int_{\mathcal{W}}\sum_{\theta}\pi(\theta)\left[u(c(\theta,\mathcal{W}),n(\theta,\mathcal{W}))+\beta V(k',H')\right]dH(\mathcal{W}) \tag{45}$$

subject to

$$\int_{\mathcal{W}}\sum_{\theta}\pi(\theta)c(\theta,\mathcal{W})dH(\mathcal{W})+g+k'-(1-\delta)k\le F\left(k,\int_{\mathcal{W}}\sum_{\theta}\pi(\theta)\theta n(\theta,\mathcal{W})dH(\mathcal{W})\right) \tag{46}$$

and

$$\sum_{\theta}\pi(\theta)\left[\beta\mathcal{W}'+u_c c(\theta,\mathcal{W})+u_n n(\theta,\mathcal{W})\right]=\mathcal{W}, \tag{47}$$

[13] Note that we assume that agents trade only assets that are contingent on their own histories. This assumption is without loss of generality. We could have allowed agents to trade assets contingent on other agents' histories without changing our results.

where the distribution H' is induced by the policy functions as follows:

$$H'\left(\mathcal{W}'\right) = \int_{A(\theta,\mathcal{W})} \sum_{\theta} \pi(\theta) dH(\mathcal{W}),$$

where $A(\theta,\mathcal{W}) = \left\{\theta, \mathcal{W} : g(\theta,\mathcal{W}) \leq \mathcal{W}'\right\}$.

We now turn to the incomplete markets case. Here, agents can only trade a risk-free bond, as in Aiyagari (1995). To retain the spirit of incomplete markets, we assume that the tax rates cannot depend on the history of shocks. If tax rates could depend on the history of shocks, then the government would effectively complete markets and our earlier analysis would apply. For simplicity, we assume that the government has access to taxes only on consumption and labor income. Then the sequence of budget constraints is

$$(1+\tau_t^c)c_t\left(\theta^t\right) - (1-\tau_t^n)\theta w_t n_t\left(\theta^t\right) + q_{t+1,t}W_{t+1}\left(\theta^t\right) = W_t\left(\theta^{t-1}\right),$$

where $W_t\left(\theta^{t-1}\right)$ is the wealth measured in (before-tax) consumption units of an agent at the beginning of period t.

Using the marginal conditions of households, we obtain the following implementability conditions:

$$\sum_{\theta} \pi\left(\theta^t\right) u_{c,t}\left(\theta^t\right) c_t\left(\theta^t\right) + \sum_{\theta} \pi\left(\theta^t\right) u_{n,t}\left(\theta^t\right) n_t\left(\theta^t\right) + \sum_{\theta} \beta \mathcal{W}_{t+1}\left(\theta^t\right) = \mathcal{W}_t\left(\theta^{t-1}\right).$$

Market incompleteness imposes additional restrictions. To develop them, it is useful to define the *ex post promised wealth* $\tilde{\mathcal{W}}_t(\theta^t)$ by

$$\tilde{\mathcal{W}}_t(\theta^t) = \frac{u_c\left(\theta^t\right) W_t\left(\theta^{t-1}\right)}{\left(1+\tau_t^c\right)}$$

for given any level of wealth $W_t\left(\theta^{t-1}\right)$. Since agents can only trade a risk-free bond, $W_t\left(\theta^{t-1}\right)$ cannot depend on the current realization of θ. Thus, an equilibrium must satisfy the following measurability restriction:

$$\frac{\tilde{\mathcal{W}}_t(\theta^{t-1},\theta)}{u_c(\theta^{t-1},\theta)} = \frac{\tilde{\mathcal{W}}_t(\theta^{t-1},\hat{\theta})}{u_c(\theta^{t-1},\hat{\theta})} \text{ for all } \theta,\ \hat{\theta},\ \theta^{t-1},$$

where ex ante and ex post promised wealth levels are related by

$$\mathcal{W}_t(\theta^{t-1}) = \sum_{\theta} \pi(\theta) \tilde{\mathcal{W}}_t(\theta^{t-1},\theta).$$

The requirement that tax rates be the same for all agents imposes restrictions as well. To develop them, we define inherited marginal utility by $\mathcal{U}_t = u_{c,t-1}\left(\theta^{t-1}\right)$. The requirement that the rate of return on the risk-free bond be the same for all agents can then be stated as

$$\frac{\sum_{\theta} u_{c,t}\left(\theta^{t-1},\theta\right)}{\mathcal{U}_t\left(\theta^{t-1}\right)} = \frac{\sum_{\theta} u_{c,t}\left(\hat{\theta}^{t-1},\theta\right)}{\mathcal{U}_t\left(\hat{\theta}^{t-1}\right)} \text{ for all } \theta^{t-1} \text{ and } \hat{\theta}^{t-1}.$$

The requirement that consumption and labor tax rates be the same for all agents can be stated as

$$\frac{u_{c,t}\left(\theta^t\right)\theta}{u_{n,t}\left(\theta^t\right)} = \frac{u_{c,t}\left(\hat{\theta}^t\right)\hat{\theta}}{u_{n,t}\left(\hat{\theta}^t\right)} \text{ for all } \theta^t \text{ and } \hat{\theta}^t.$$

The aggregate state variables are k and the joint distribution H over ex ante promised wealth $\mathcal{W}$ and inherited marginal utility $\mathcal{U}$. The recursive formulation of the Ramsey problem is to choose $c(\theta,\mathcal{W},\mathcal{U})$, $n(\theta,\mathcal{W},\mathcal{U})$, k', and the policy functions for future promised wealth $\mathcal{W}' = g(\theta,\mathcal{W},\mathcal{U})$, and marginal utility $\mathcal{U}' = h(\theta,\mathcal{W},\mathcal{U})$, and the implied next-period joint distribution H' to solve

$$V(k,H) = \text{Max} \int_{\mathcal{W},\mathcal{U}} \sum_{\theta} \pi(\theta)\left[u(c(\theta,\mathcal{W},\mathcal{U}), n(\theta,\mathcal{W},\mathcal{U})) + \beta V(k',H')\right] dH(\mathcal{W},\mathcal{U}) \tag{48}$$

subject to the resource constraint (46), the implementability condition (47), the constraints that bond rates must be the same for all agents,

$$\frac{\sum_{\theta} u_c(\theta,\mathcal{W},\mathcal{U})}{\mathcal{U}} = \frac{\sum_{\theta} u_c\left(\theta,\widehat{\mathcal{W}},\widehat{\mathcal{U}}\right)}{\widehat{\mathcal{U}}}, \text{ for all } \mathcal{W},\mathcal{U} \text{ and } \widehat{\mathcal{W}},\ \widehat{\mathcal{U}}, \tag{49}$$

the constraints that tax rates must be the same for all agents,

$$\frac{u_c(\theta,\mathcal{W},\mathcal{U})\theta}{u_n(\theta,\mathcal{W},\mathcal{U})} = \frac{u_c\left(\hat{\theta},\widehat{\mathcal{W}},\widehat{\mathcal{U}}\right)\hat{\theta}}{u_n\left(\hat{\theta},\widehat{\mathcal{W}},\widehat{\mathcal{U}}\right)} \text{ for all } \theta,\mathcal{W},\mathcal{U} \text{ and } \hat{\theta},\ \widehat{\mathcal{W}},\ \widehat{\mathcal{U}}, \tag{50}$$

and the incomplete markets constraints

$$\frac{\tilde{\mathcal{W}}(\theta,\mathcal{W},\mathcal{U})}{u_c(\theta,\mathcal{W},\mathcal{U})}=\frac{\tilde{\mathcal{W}}\left(\hat{\theta},\widehat{\mathcal{W}},\widehat{\mathcal{U}}\right)}{u_c\left(\hat{\theta},\widehat{\mathcal{W}},\widehat{\mathcal{U}}\right)}\text{ for all }\theta,\mathcal{W},\mathcal{U}\text{ and }\hat{\theta},\ \widehat{\mathcal{W}},\ \widehat{\mathcal{U}},\tag{51}$$

where $\mathcal{W}=\int_{\mathcal{U}}\sum_{\theta}\pi(\theta)\tilde{\mathcal{W}}(\theta,\mathcal{W},\mathcal{U})dH(\mathcal{W},\mathcal{U})$, and where inherited marginal utility evolves according to

$$\mathcal{U}'=u_c(\theta,\mathcal{W},\mathcal{U}).\tag{52}$$

The distribution H' is induced by the policy functions as follows:

$$H'\left(\mathcal{W}',\mathcal{U}'\right)=\int_{A(\theta,\mathcal{W},\mathcal{U})}\sum_{\theta}\pi(\theta)dH(\mathcal{W},\mathcal{U})$$

where

$$A(\theta,\mathcal{W},\mathcal{U})=\left\{\theta,\mathcal{W},\mathcal{U}:g(\theta,\mathcal{W},\mathcal{U})\leq\mathcal{W}'\text{ and }h(\theta,\mathcal{W},\mathcal{U})\leq\mathcal{U}'\right\}.$$

With the added constraints implied by incomplete markets, the logic of the proof of Proposition 3 does not go through. Characterizing the solution to this problem is beyond the scope of this paper, but it seems likely that the added constraints will imply that zero taxation of capital is no longer optimal.

4. Uniform taxation and production efficiency

In this section, we connect our results to the results on production efficiency and uniform taxation. We obtain these connections by recasting our dynamic economies into equivalent static economies. With this recasting, we show that Proposition 3 regarding the optimality of zero taxation of capital is an implication of production efficiency in the static economy. We then show that the first part of Proposition 6, which asserts that, if preferences are the same for the two agents, zero taxation of capital is optimal, is an implication of production efficiency in the recast economy. If, instead, preferences are different, it turns out that the recast economy has an additional restriction on tax instruments, and production efficiency is no longer optimal. Thus, the recasting helps to shed light on the link between zero taxation of capital and production efficiency.

4.1. Production efficiency in a representative agent economy

Consider an alternative economy, which we call an *intermediate goods economy*, that seems different at face value but turns out to be equivalent to the one considered above. In this alternative economy, the representative household consumes a single final good denoted by C and supplies a single final labor input denoted by N. Preferences for the households are given by

$$U(C,N)=\frac{C^{1-\sigma}-1}{1-\sigma}-\eta N^{1+\psi}.\tag{53}$$

The economy has three types of technologies. The first one is given by the resource constraint (2). We refer to the representative firm that operates this technology as the *capital accumulation firm*. This firm produces intermediate goods c_t, hires intermediate labor inputs n_t, and accumulates capital according to the technology (2). The second technology, operated by the *consumption firm*, produces the final good C using the intermediate goods c_t according to the constant returns to scale technology given by

$$C=\mathcal{C}(c_0,c_1,\ldots)=\left[\sum_{t=0}^{\infty}\beta^t c_t^{1-\sigma}\right]^{\frac{1}{1-\sigma}}.\tag{54}$$

The third type of technology, operated by the *labor firm*, produces the intermediate labor inputs using final labor according to the constant returns to scale technology given by

$$N=\mathcal{N}(n_0,n_1,\ldots)=\left[\sum_{t=0}^{\infty}\beta^t n_t^{1+\psi}\right]^{\frac{1}{1+\psi}}.\tag{55}$$

In terms of the tax system, we assume that the government can levy a tax on the final consumption good, τ^c, and on final labor denoted by τ^n. In addition, we assume that the government can levy taxes on all intermediate goods, denoted by τ_t^c and τ_t^n. We retain the dividend taxes and the capital income taxes levied on the capital accumulation firm, as well as the initial levy l_0. We do not impose any taxes on the profits of either the consumption firm or the labor firm because these profits will be zero in equilibrium.

The households' problem is to maximize (53) subject to the budget constraint

$$p(1+\tau^C)C - w(1-\tau^N)N \leq (1-l_0)V_0, \tag{56}$$

where p and w denote the prices of final consumption and labor in units of the consumption good in period zero, and V_0 is the value of initial wealth in units of goods.

The consumption firm's problem is to maximize

$$pC - \sum_{t=0}^{\infty} q_t(1+\tau_t^c)c_t \tag{57}$$

subject to (54), and the labor firm's problem is to maximize

$$\sum_{t=0}^{\infty} q_t(1-\tau_t^n)w_t n_t - wN \tag{58}$$

subject to (55).

The capital accumulation firm's problem is the same as in our benchmark model. A competitive equilibrium is defined in the standard fashion.

Next, we show that the competitive equilibria in the two economies coincide in the intermediate goods economy and in the growth economy with distorting taxes. To do so, we first show that we can rewrite the equilibrium in the intermediate goods economy as an equilibrium in the growth economy with taxes by directly incorporating the decisions of the consumption firm and the labor firm into the households' problem.

Consider the budget constraint of the households. Given that in any competitive equilibrium, profits are zero for the consumption firm and the labor firm, we can substitute $pC = \sum_{t=0}^{\infty} q_t(1+\tau_t^c)c_t$ from (57) and $wN = \sum_{t=0}^{\infty} q_t(1-\tau_t^n)w_t n_t$ from (58) into (56) to obtain a budget constraint of the form (11) in the growth economy with taxes. The only difference is the presence of the tax on final consumption and final labor, which amounts to rescaling the consumption and labor income taxes in the original economy.

Substituting from (55) and (54) into (53), we see that the households' utility function in the rewritten intermediate goods economy is the same as in the growth economy with taxes.

To establish that the converse holds, note that we can set up the households' problem in the growth economy with taxes as a two-stage problem of first choosing an aggregate value for consumption and labor and then choosing the disaggregated levels of consumption and labor to achieve the desired value for final consumption and labor. Thus, the equilibria in the two economies coincide.

In the intermediate goods economy, the Ramsey problem is to maximize (53) subject to the implementability constraint

$$U_C C + U_N N = \bar{\mathcal{W}}, \tag{59}$$

where $\bar{\mathcal{W}}$ denotes the exogenously specified bound on the value of wealth the planner must deliver and is given by

$$\bar{\mathcal{W}} = \frac{U_C}{1+\tau^C}(1-l_0)V_0$$

and the requirement that the allocation is in the *production set* given by (2), (54) and (55) with inequalities.

Next, when we apply the same logic as in Diamond and Mirrlees (1971) to our intermediate goods economy, it is immediate that the solution to the Ramsey problem must satisfy production efficiency. An implication of this result is that implementing the Ramsey equilibrium with no taxation of intermediate goods is possible.

Thus, setting these taxes to zero, we can combine the first-order conditions for the capital accumulation firm and the consumption firm to obtain

$$\frac{\mathcal{C}_{ct}}{\mathcal{C}_{ct+1}} = 1 - \delta + F_{k,t+1}. \tag{60}$$

Similarly, combining the first-order conditions for the capital accumulation firm and the labor firm, we obtain

$$\frac{\mathcal{N}_{nt}}{\mathcal{N}_{nt+1}} = \frac{F_{n,t}}{F_{n,t+1}}\left(1 - \delta + F_{k,t+1}\right). \tag{61}$$

Condition (60) equates the rates at which c_t is transformed into c_{t+1} through the composite C to the rate at which c_t is transformed into c_{t+1} through capital, k_{t+1}. Condition (61) is the analog for labor in consecutive periods. Note that

$$\frac{\mathcal{C}_{ct}}{\mathcal{C}_{ct+1}} = \frac{c_t^{-\sigma}}{\beta c_{t+1}^{-\sigma}} = \frac{u_{c,t}}{\beta u_{c,t+1}}. \tag{62}$$

A similar expression holds for the intermediate labor inputs.

The intermediate goods economy and our benchmark economy are obviously equivalent. This observation, together with (60) and (62), as well as the analogous conditions for labor, implies that not taxing capital in the benchmark economy is optimal. We summarize this discussion in the following proposition, which is the equivalent of Proposition 3 in the intermediate goods economy.

Proposition 3'. *(No intertemporal distortions ever) Suppose that preferences and technologies are given by (53) , (54), and (55), and the wealth restriction must be satisfied. Then, the Ramsey solution has no intertemporal distortions for all* $t \geq 0$.

Consider next the case in which rather than imposing a wealth constraint, initial policies are restricted so that rents cannot be fully taxed away. Here, because rents are not fully taxed away, production efficiency is no longer optimal. Nevertheless, it is straightforward to show that an analog of Proposition 5 holds in the intermediate goods economy in the sense that distorting intermediate goods decisions from period one onward is not optimal. Distorting intermediate goods in period zero relative to all other intermediate goods is optimal in order to partially tax rents.

The intermediate goods economy helps to clarify circumstances in which not having intertemporal distortions is optimal. We have shown that it is optimal to not have such distortions when the underlying economy can be represented as a constant returns to scale economy in the production of intermediate goods. In this sense, we have shown an equivalence between the principle that intermediate goods taxation is undesirable and intertemporal distortions should not be introduced.

In the process of doing so, we have shown that the celebrated result of Atkinson and Stiglitz (1972) – that is, that uniform commodity taxation is optimal when preferences are homothetic and separable – follows from the production efficiency result of Diamond and Mirrlees (1971). Thus, our result of no intertemporal distortions is very closely connected to the uniform commodity taxation result.

4.2. Production efficiency in a heterogeneous agents economy

Now consider developing an intermediate goods economy that is equivalent to our heterogeneous agents economy. Here we think of the intermediate goods economy as producing two distinct types of final consumption goods, denoted by C^i for $i = 1, 2$. For simplicity, we assume the economy utilizes one common type of final labor, denoted by N. The preferences for households of type i are given by

$$U\left(C^i, N^i\right) = \frac{\left(C^i\right)^{1-\sigma^i} - 1}{1-\sigma^i} - \eta\left(N^i\right)^{1+\psi}, \tag{63}$$

where N^i denotes the amount of the common final labor supplied by type i. The technologies for the capital accumulation firm and the labor firm are the same as before. Each consumption good is produced by its own constant returns to scale technology, given by

$$C^i = C^i(c_0^i, c_1^i, \ldots) = \left[\sum_{t=0}^{\infty} \beta^t \left(c_t^i\right)^{1-\sigma^i}\right]^{\frac{1}{1-\sigma^i}} \tag{64}$$

for $i = 1, 2$.

The tax system is the same, except that now we require that the tax rate on final consumption goods must be the same for the two types.

Households of type i maximize (63) subject to the budget constraint

$$p^i(1+\tau^C)C^i - w(1-\tau^N)N^i \leq (1-l_0)V_0^i, \tag{65}$$

where p^i denotes the price of the final consumption good of type i in units of the consumption good in period zero.

The consumption firm of type i maximizes

$$p^i C^i - \sum_{t=0}^{\infty} q_t(1+\tau_t^c)c_t^i \tag{66}$$

subject to (64).

The other firms solve the same problems as in the representative agent economy. A competitive equilibrium is defined in the standard fashion.

Next, we show that the competitive equilibria in the intermediate goods economy and the heterogeneous agents economy with type-independent taxes coincide. Recall that in the representative agent model, we used zero profits for consumption goods producers and replaced the pretax value of consumption in the households' budget constraint in the intermediate goods economy with the value of the intermediate goods. Following the same procedure, we can use (66) and write (65) as

$$(1+\tau^C)\sum_{t=0}^{\infty} q_t(1+\tau_t^c)c_t^i - w(1-\tau^N)N^i \leq (1-l_0)V_0^i.$$

Notice that this budget constraint coincides with the budget constraints in the heterogeneous agents economy except for rescaling. The rest of the argument that the equilibria coincide is the same as in the representative agent economy.

Now consider the Ramsey problem in the intermediate goods economy. Since the tax rates on both consumption goods must be the same, the Ramsey problem has an additional restriction that can be written as

$$\frac{U^i_{C^i}\mathcal{C}^i_{c_t}}{U^i_{N^i}\mathcal{N}^i_{n_t}} = \frac{U^j_{C^i}\mathcal{C}^j_{c_t}}{U^j_{N^i}\mathcal{N}^j_{n_t}},\ t \geq 0. \tag{67}$$

The Ramsey problem is now to maximize utility subject to the implementability constraint, the requirement that the allocation is in the production set and (67). The Ramsey problem in the intermediate goods economy with type-dependent taxes simply drops (67).

The restriction (67) implies that the Ramsey outcomes are not, in general, production efficient. In the case in which the preferences of the two agents are the same, the outcomes with type-dependent taxes coincide with those with type-independent taxes, so that (67) is not binding.

We now have the analog of Proposition 6.

Proposition 6'. *(No intertemporal distortions in heterogeneous agents economies) Suppose that preferences for all types of agents are in the class of standard macroeconomic preferences. If all agents have the same preferences, then the Ramsey equilibrium has no intertemporal distortions for all $t \geq 0$. If $\sigma^1 \neq \sigma^2$, the Ramsey equilibrium has intertemporal distortions.*

Remark. Propositions 6 and 6′ help to clarify the relationship between zero taxation of capital and the optimality of production efficiency emphasized by Diamond and Mirrlees (1971). If all agents have the same preferences, then zero taxation of capital follows from production efficiency in the recast economy. If preferences differ and tax rates cannot be type-specific, then zero taxation of capital is not optimal and production efficiency fails in the recast economy. If preferences differ but tax rates can be type-specific, the optimality of zero taxation of capital follows from the optimality of production efficiency.

5. Concluding remarks

We have extended the production efficiency theorem to environments in which taxation of pure rents is limited but the Ramsey planner faces a wealth constraint. Our notion of zero taxation of capital is that there are no intertemporal distortions. This notion is stronger than production efficiency but follows from it for the kinds of preferences that are standard in the macroeconomics literature. Using these two results, we have shown that in standard macroeconomic models, production efficiency requires that capital taxation is inefficient if the planner has access to a rich tax system.

We have also argued that setting up Ramsey problems with wealth constraints has the desirable feature that, with partial commitment, the Ramsey outcomes are time consistent.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jmoneco.2019.09.015.

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